Edge Cases - Voice Assistant on ESP32

Group A2, IoT course, Summer 2026.

Cases the system knows how to handle, and how to see them in the demo.

#	The case	What the system does	How to demo it	✓
1	Noise in the room in Hands-free mode (a knock on the table, a clap, footsteps)	The voice detector recognizes that it is not speech and does not start recording.	Knock on the table and clap next to the device. The state stays Waiting for speech.	
2	Silence	In Hands-free mode, silence is not sent to the cloud at all. When using the button, the system answers with a generic sentence.	Hands-free on and stay silent: no recording starts. Click Talk and stay silent: the assistant answers with a generic sentence.	
3	Clicking Talk while the assistant is still answering	A second recording cannot start while the assistant is speaking (also a greeting, a timer message or an answer to a typed question): the request is rejected (also `LISTEN` from the terminal and BOOT, with a message). Wait for it to finish, or click Cancel.	In the middle of an answer the Talk button shows Busy and cannot be clicked, and the space bar does not start a recording either.	
4	A short pause in the middle of a sentence	The recording continues until about a second of silence (can be changed with Stop after silence).	Say a sentence with a short stop in the middle. It is sent as one question.	
5	The connection to OpenAI drops	The board reconnects on its own, with a growing wait between attempts (backoff). After 5 failed attempts in a row it goes Offline (red OpenAI light) and keeps retrying every 30 seconds, never stopping. When the network comes back it reconnects within a few seconds, and when the connection comes back it recovers on its own.	Turn off the hotspot for a few seconds and turn it back on. The OpenAI light turns red, and after the network comes back it turns green again on its own.	
6	Board Reset	The settings (volume, mute, brightness, light thresholds, Hands-free) are saved in flash and come back after startup.	Change the volume, press Reset, and check that the volume stayed.	
7	The control center disconnects in the middle	A red banner appears, and when you connect again the page reconnects on its own and gets the current state (STATUS).	Click Disconnect and then Connect, or unplug the USB cable and plug it in again. All the lights and the state appear again.	
8	Volume out of range	The board limits the volume to 0 to 100.	Say `Set the volume to 150 percent`. The volume stops at 100%.	
9	Speech from a standard distance (a meter) with background noise	The voice detector recognizes the speech even with noise in the background, and the assistant understands the request and carries it out.	One person drums on the table, and the other stands a meter from the device and says `Turn on the light`. The LED turns on.	
10	The timer ends when there is no connection to OpenAI	The timer runs on the board itself, so it beeps and blinks even without internet. Only the spoken message needs a connection: if there is no connection when the timer ends, it is not said.	Start `Set a timer for 30 seconds` and turn off the hotspot. After 30 seconds the board beeps.	
11	Several hardware requests in one command	The assistant catches all the requests in the command and carries them all out together.	Say `Set the light brightness and the volume to 100 percent`. Both the LED brightness and the volume go up to 100%.	